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ORIGINAL ARTICLE

LieWaves: dataset for lie detection based on EEG signals and wavelets

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Abstract

This study introduces an electroencephalography (EEG)-based dataset to analyze lie detection. Various analyses or detections can be performed using EEG signals. Lie detection using EEG data has recently become a significant topic. In every aspect of life, people find the need to tell lies to each other. While lies told daily may not have significant societal impacts, lie detection becomes crucial in legal, security, job interviews, or situations that could affect the community. This study aims to obtain EEG signals for lie detection, create a dataset, and analyze this dataset using signal processing techniques and deep learning methods. EEG signals were acquired from 27 individuals using a wearable EEG device called Emotiv Insight with 5 channels (AF3, T7, Pz, T8, AF4). Each person took part in two trials: one where they were honest and another where they were deceitful. During each experiment, participants evaluated beads they saw before the experiment and stole from them in front of a video clip. This study consisted of four stages. In the first stage, the LieWaves dataset was created with the EEG data obtained during these experiments. In the second stage, preprocessing was carried out. In this stage, the automatic and tunable artifact removal (ATAR) algorithm was applied to remove the artifacts from the EEG signals. Later, the overlapping sliding window (OSW) method was used for data augmentation. In the third stage, feature extraction was performed. To achieve this, EEG signals were analyzed by combining discrete wavelet transform (DWT) and fast Fourier transform (FFT) including statistical methods (SM). In the last stage, each obtained feature vector was classified separately using Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and CNNLSTM hybrid algorithms. At the study's conclusion, the most accurate result, achieving a 99.88% accuracy score, was produced using the LSTM and DWT techniques. With this study, a new data set was introduced to the literature, and it was aimed to eliminate the deficiencies in this field with this data set. Evaluation results obtained from the data set have shown that this data set can be effective in this field.

Keywords EEG · Lie detection · CNN · LSTM · DWT · FFT · ATAR · OSW

1 Introduction

EEG signals provide valuable insights into various aspects of human behavior, enabling advancements that can simplify human life. EEG involves recording electrical activity across the scalp and displaying this activity on paper or a computer. It can track brain waves containing the physiological and functional details of brain activities, aiding in understanding the brain's intricate workings [1, 2]. Numerous studies have been conducted on EEG signals, primarily focusing on addiction detection [3, 4], emotion analysis [5–7], sleep state detection [8, 9], identity verification [10–12], device control [13, 14], and, more recently, lie detection [23–25].

Lying is a fundamental aspect of human nature. Individuals may resort to lying to justify themselves or escape from wrongdoing. Such situations are more prevalent, especially in security and criminology, where lie detection is often vital [15]. Traditional lie detection methods, such as verbal questioning, written surveys, and polygraphs, are subjective and time-consuming. The validity of devices like magnetic resonance imaging (MRI) and polygraphs has been criticized due to potential intentional deception by individuals familiar with the devices [16]. Following the development of polygraph tests, functional magnetic resonance imaging (fMRI) gained popularity in lie detection. However, its high cost,

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technological complexity, and the challenge of predicting outcomes questioned the validity of these approaches [17]. For these reasons, new techniques have been continually developed in response to questioning the validity of these methods. Studies are now focusing on processing facial expressions and audio data from video recordings using deep learning methods to classify whether individuals in the video are telling the truth or lying [18–22]. In addition to these studies, using EEG signals for lie detection has recently become a prominent topic [23–25]. It is a known fact that when a person lies, the strength of the signal in the brain increases [26]. If the signal surpasses a certain threshold, it can be detected that the person is lying at that moment [26]. Although a trained individual can maintain composure and deceive specific methods, suppressing mental activities in the brain is challenging [24].

Using EEG signals for lie detection aims to overcome the disadvantages of traditional methods. Like many methods, EEG signals also have disadvantages. Modern EEG devices are sensitive instruments, and great care must be taken during the data collection phase. The environment where signals are collected should be protected as possible from external factors. More reliable results can be obtained from signals collected in such an environment. However, recently developed EEG devices are becoming more suitable for everyday use. Additionally, with advancing technology, noise reduction algorithms have successfully removed signal artifacts. As user-friendly devices become more prevalent and improve, the disadvantages associated with lie detection from EEG signals will diminish over time. The success of lie detection using EEG signals and increasing accuracy in this field will benefit humanity in many areas. It can help prevent lying in various contexts, including legal cases, job interviews, trade, and management. Especially in legal cases, lies can affect many people. There are many examples of innocent people spending years in prison and guilty people escaping the punishment they deserved. In such cases, the importance of lie detection becomes even more significant.

For these reasons, lie detection based on EEG signals was carried out in this study. This study aims to introduce the LieWaves dataset, created by obtaining EEG signals from different individuals for lie detection, and to design a model that processes and classifies this dataset using signal processing techniques and deep learning methods to determine whether individuals are telling the truth or lying. The goal is to overcome the disadvantages of traditional lie detection methods in legal cases or situations where lie detection is necessary and achieve better results using different datasets and methods compared to previous EEG-based lie detection studies. The proposed study demonstrates that lie detection using EEG signals is possible and can yield successful results. As mentioned earlier, the motivation for this study is to use EEG signals as a novel method for lie detection and to increase the accuracy of lie detection. As discussed, experienced or composed individuals can professionally deceive many lie detection methods. This study shows the strength of EEG signals in lie detection area. The deep learning model developed using the LieWaves dataset in this study could be further translated into a practical application. Implementing a lie detection system based on this model makes applicable use in areas where lie detection is crucial. The LieWaves dataset created in this study can be included in the literature, providing an opportunity to develop different studies or applications that may yield better results.

The LieWaves dataset was collected from 27 subjects using a 5-channel (AF3, T7, Pz, T8, AF4) Emotiv Insight device. Each subject participated in two experiments, one involving telling the truth and the other involving lying. Visual stimuli were presented to the subjects during these experiments, and different beads were used as visual stimuli. EEG data were collected from each subject during each experiment, creating an EEG dataset. This study introduces and analyzes the LieWaves dataset for lie detection. EEG data analysis involves four stages, as shown in Fig. 1. The first stage is the acquisition of EEG data. The second stage involves preprocessing the obtained signals, addressing artifact removal, and signal filtering. The third stage is feature extraction, where specific features are extracted from EEG data for more accurate dataset analysis. The fourth stage involves classifying the extracted features using deep learning methods and evaluating the model.

In the scope of this study, EEG data were filtered through a bandpass filter in the range of 0.5–45 Hz. Subsequently, the ATAR algorithm was applied to remove noise from these EEG data. Noise-reduced data underwent data augmentation using the OSW method. Later, the DWT and FFT methods were employed separately, and then statistical methods were considered individually for each method to create feature vectors. The obtained feature vectors were used in the classification process with CNN, LSTM, and the combination of both methods, CNNLSTM, to create models. When the models were evaluated, the best performance was achieved

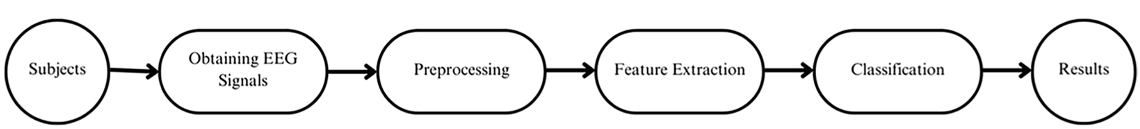


Fig. 1 Stages of analyzing EEG signals

by applying DWT and LSTM methods, yielding an accuracy rate of 99.88%. Considering all results, it was observed that the obtained outcomes demonstrated high levels of success. Compared to previous studies on lie detection, the results of this study were found to be satisfactory. Key highlights of the research can be summarized as follows:

- An original dataset, LieWaves, was created for lie detection using EEG data.
- This study introduced a dataset obtained with a 5-channel EEG device (AF3, T7, Pz, T8, AF4), which distinguishes it from existing datasets.
- This study achieved better results with fewer electrodes in lie detection.
- The created dataset was processed with the ATAR algorithm to eliminate artifacts, a method not commonly used before.
- While studies using CNN and LSTM methods for the classification of EEG signals in lie detection exist, to the best of our knowledge, there is no study using a hybrid combination of both methods, such as CNNLSTM.

The continuation of the study is organized as follows: In the second section, some studies related to lie detection are discussed, and evaluations of these studies are presented. The third section covers EEG datasets used for lie detection in the literature. The fourth section discusses the steps taken to obtain the dataset and the methods and techniques used for dataset analysis. The fifth section provides the study's results, and the findings concerning the literature are discussed. The advantages and disadvantages of the study are also addressed in this section. The last section offers a general summary of the study and outlines potential future research directions.

2 Related works

Several studies have been conducted on deception detection using EEG signals. Due to its relatively recent emergence as a topic of interest, there are few studies on this subject. Additionally, the available datasets for these studies are limited. Some of the studies on deception detection using EEG signals are as follows:

In the study [23], researchers conducted a study on deception detection based on P300 waves using EEG signals. P300 signals are signals with a positive peak between 300 and 1000 ms from the onset of the stimulus. The study utilized the Dryad.2qc64 dataset was created with EEG data obtained from 30 participants using a 12-channel EEG device. EEG signals were preprocessed using independent component analysis (ICA), signal-to-noise ratio (SNR), and spatial de-noising algorithm (SDA) methods. Three feature groups were extracted using these methods, and the most suitable features were selected using the F-score method. Support vector machine (SVM) was used for classification, achieving an accuracy rate of 96.05% at the end of the study. This study focused on P300 waves and demonstrated successful deception detection from EEG signals using traditional machine learning methods with a high accuracy rate.

Study [27] conducted a deception detection study using EEG signals and the P300 wave. EEG data were obtained from 12 participants using a 5-channel EEG device. Discrete wavelet transform (DWT) was employed for feature extraction, and adaptive neuro-fuzzy inference system (ANFIS) was used for classification. The study achieved a discrimination accuracy of 64.27% between truth-tellers and liars using P300 waves. Although P300 waves were used for deception detection, the study obtained a relatively low accuracy rate. Possible reasons for the low performance include insufficient data and a need for more alignment between the applied methods and the data.

In study [28], a study was conducted to demonstrate that deception detection using EEG signals is more successful than traditional lie detector tests. The study included ten participants, and data were recorded with a 16-channel EEG device while subjects were shown specific images. Feature extraction from EEG data was performed using empirical mode decomposition (EMD). The feature vector was then classified using support vector machine (SVM), K-nearest neighbors (KNN), quadratic discriminant analysis (QDA), and decision tree (DT) methods. The SVM method achieved the best classification results, reaching an average accuracy of 81.71%. While this study highlighted the potential of deception detection using EEG signals, there is room for improvement, and higher success rates could be achieved with diverse data and different methods.

In the study [29], EEG signals from a 9-channel EEG device were used for deception detection, with data obtained from 33 participants. Feature extraction from EEG signals was performed using the empirical mode decomposition (EMD) method, and SVM was used for the classification process. The study achieved a classification accuracy of 99.44%. This study demonstrated the high potential of deception detection using EEG signals, but the methods should be evaluated on different datasets to strengthen the results.

A study [30] proposed a scheme for P300-based deception detection. EEG records were obtained from 15 participants during deception detection, and independent component analysis (ICA) was applied to extract features from the EEG records. Subsequently, SVM, linear discriminant analysis (LDA), K-nearest neighbors (KNN), and backpropagation neural network (BPNN) methods were used for classification. The classification results yielded 74.5%, 79.4%, 97.9%, and 89% success rates, respectively. This study showed promising results for deception detection using P300 waves, and the comparison of different classification algorithms enhanced the reliability of the study. However, further improvement could be achieved by diversifying the data and increasing the number of subjects.

Study [31] conducted a deception detection study using an EEG headset to analyze brainwave patterns. Single-channel EEG signals were obtained during question–answer sessions, and these signals were classified using SVM, decision forest (DF), and neural network (NN) algorithms. The study achieved success rates of 63.1%, 99.2%, and 55.7%, respectively. While this study demonstrated the possibility of deception detection using EEG signals, the high discrepancy between classifiers may affect the reliability of the study. Evaluating the results by comparing the same methods on different datasets would be appropriate.

In study [19], a dataset named Bag-Of-Lies was created for deception detection using EEG signals, eye tracking, video footage, and audio data. Participants were asked to lie or tell the truth about visuals shown during dataset creation, and EEG data were recorded. Random forest (RF), EEGNet, and multilayer perceptron (MLP) methods were applied to EEG data for classification. The highest accuracy score, 58.71%, was achieved with the RF method. Although the dataset created in this study serves as a valuable resource for deception detection using EEG signals, the low results are attributed to the absence of signal processing techniques.

In study [24], a model was developed using a Convolutional Neural Network (CNN) based on 12-channel EEG signals obtained from 30 participants to determine if a person is telling the truth or lying. The model achieved a test accuracy of 84.44%. While this study obtained good results in deception detection from EEG signals using deep learning methods, there is potential for improvement by applying different data or signal processing techniques.

The study [22] applied three different deception and lie detection methods. A study on deception detection was conducted by classifying visual, auditory, and EEG signals from individuals using deep learning methods. Facial movements were classified using CNN, and the same approach was applied to create a lie detection system for sound signals. A bidirectional Long Short-Term Memory (LSTM) algorithm was used for deception detection with EEG signals. The three methods were combined to create a model for deception detection. Experimental results showed a success rate of 83.5%. The study is significant for performing deception detection in parameters beyond EEG. Increasing the number of data samples and applying different signal and image processing techniques are suggested to improve the results obtained.

In study [20], a Deep Convolutional Neural Network (DCNN) method was proposed for a multi-purpose deception detection model named LieNet. The study used the Bag-Of-Lies, Real-life (RL) trail, and Miami University Deception Detection (MU3D) datasets. For the Bag-Of-Lies dataset, 20 frames were taken from each video to create a single image and audio signals were included in the image. A 13-channel EEG signal was also combined with the image by drawing it in 2D. DCNN models were applied to each method. For the Bag-Of-Lies dataset, average accuracy rates of 95.91% and 96.04% were achieved for SetA and SetB, respectively. RL and MU3D datasets achieved 97% and 98% success rates, respectively. These results demonstrate high accuracy by transforming EEG signals into images and applying a 2D-CNN model.

All these studies collectively demonstrate that deception detection using EEG signals is feasible, and high accuracy rates can be achieved. While some studies obtained low performance, overall, high accuracy values were obtained. Improvements can be made by increasing the number of data samples and applying different signal processing techniques and appropriate classifiers. The datasets and methods used in studies on deception detection using EEG signals provided valuable insights during this study's preparation and implementation stages. In contrast to the procedures for obtaining EEG data in existing studies, different data acquisition procedures and devices were applied to create the LieWaves dataset in this study. The primary goal of referencing these studies in the literature is to highlight this study's unique aspects and contributions to the existing body of knowledge.

3 Materials and methods

In this section, details are provided regarding the procedures involved in constructing the dataset for the study, including the stimuli employed and the equipment utilized throughout these procedures. In addition, detailed information is given about the pre-processing methods used before attempting the classification process, various signal processing methods, and feature engineering. The study consisted of four stages. In the first stage, data were obtained from 27 different subjects with EEG signals, and the data set began to be created. Then, in the second stage, the ATAR algorithm was used to eliminate the artifacts that occurred while obtaining EEG signals, and the EEG signals were cleaned. After the cleaning process, in this step, the size of the data was increased using OSW, and it was made ready for feature extraction. In the third stage, feature extraction was performed on the augmented EEG data from each EEG channel, and DWT and FFT were used separately for this. In the last stage, the classification process was carried out using CNN, LSTM, and CNNLSTM deep learning algorithms, and the results of the classifiers were measured with accuracy evaluation metrics. A visual explanation of the operations performed is given in Fig. 2. As shown in the Fig. 2, the strategy diagram consists

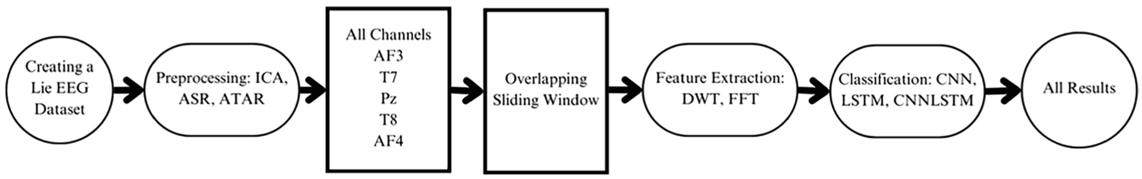


Fig. 2 Flow chart of the proposed study

of four stages. The first stage involves obtaining EEG data, during which the LieWaves dataset is created by collecting EEG data. The second stage is the preprocessing of the data. In this stage, a bandpass filter passing through the frequency range of 0.5–45 Hz is initially applied, followed by the implementation of the ATAR algorithm to remove artifacts from the EEG data. After the preprocessing stage, the data is processed for each channel individually and collectively for all channels. Subsequently, data augmentation is performed using the OSW method to obtain more reliable results. The third stage involves feature extraction, where the DWT and FFT methods are applied separately. The final stage is classification, where CNN, LSTM, and CNNLSTM algorithms are applied individually. Through these processes, different results are obtained for each method and channel. This allows for both method comparison and the attainment of channel-specific outcomes.

3.1 LieWaves dataset

3.1.1 Stimuli

A LieWaves dataset was created for the deception and truth conditions. Visual stimuli were presented to the participants during the acquisition of EEG data. In the study, ten different prayer beads were used as stimuli. Two experiments, one for deceptive and one for truthful statements, were conducted for each participant. Among the ten prayer beads used, five were used in the first experiment, and five were used in the second experiment. Figure 3 displays the visuals of the stimuli used in the study. The stimuli were presented to the participants as video clips. Each experiment showed a video clip consisting of 25 frames, with each prayer bead displayed for 2 s, with a total of five iterations. Prior to each frame, except for the first frame, a 1-s black screen was displayed. Before the first frame, a 3-s black screen was shown, but the additional 2 s in this interval were excluded from the EEG data. As a result, 75 s of EEG data were obtained, with each experiment yielding five prayer bead images and five iterations, and each presentation consisting of 1 s of the black screen followed by 2 s of the image.

3.1.2 Subjects

EEG data were collected from 27 participants, who were students and faculty members of the Faculty of Technology at Karadeniz Technical University, Of Campus. The average age of the participants was 23.1 years. They were individuals without any health problems. They were informed about the study before obtaining EEG data from the participants. The steps to be followed during the study were explained to each participant individually. In line with the ethics committee's

Fig. 3 The stimuli used in the study

approval, each participant signed a voluntary consent form, and a copy of the form was given to them. For each participant, two experiments were conducted, one as a liar and the other as a truth-teller. Different stimuli were used in each experiment, and in the first experiment, the participant was asked to decide whether to be truthful or a liar.

Before collecting the EEG data, the participant was given a box containing five prayer beads and left alone in the room. They were asked to take any two prayer beads from the box and keep them in their pockets. Then, two buttons were placed in front of the participant. The button on the right-hand side indicated "YES," and the button on the left indicated "NO." During the experiment, a video clip was projected on the wall. Before starting the experiment, the participants were instructed to close their eyes for 10 s and then open them for 10 s, ensuring their concentration. If the participant decided to play the role of a liar in the first experiment, they were instructed to press the left button indicating "NO" when they saw the image of the prayer bead they had taken and to press the right button indicating "YES" when they saw an image of a prayer bead they had not taken. In this way, the participant provided dishonest responses to all the images. As a truth-teller, the participants were asked to respond accurately to all options. That is, when they saw the image of the prayer bead they had taken, they were instructed to press the right button indicating "YES," and when they saw an image of a prayer bead they had not taken, they were instructed to press the left button indicating "NO." In this manner, each participant responded to both situations. After each experiment, the participants were given a short resting period. The study lasted approximately 30–45 min for each participant.

3.1.3 Experimental setup

The EEG data were collected under the approval of the Non-Invasive Research Ethics Committee of Firat University. The Emotiv Insight device, which has 5 channels (AF3, T7, Pz, T8, AF4), was used to obtain the EEG data. This device was chosen because similar studies did not utilize EEG data collected with this device, and a lower number of channels were preferred to achieve better results. The device is wireless and portable, providing convenience in data collection. The internal sampling rate of the device, which is 2048 Hz, was sampled at 128 Hz. Digital notch filters at 0.5–45 Hz, 50 Hz, and 60 Hz were applied within the device. Additionally, a built-in digital 5th-order sinc filter was applied [32].

The experiments were conducted in a quiet room free from noise and interference that could affect the signal waves. The stimuli were projected onto the wall using a projection device, ensuring that the participants sitting 2–3 m away could clearly see the stimuli. The EEG data were collected using the Emotiv Pro software that is compatible with the device. The device configuration was performed using this software after attaching the EEG device to the participants' heads. The electrodes of the EEG device were placed according to the 10–20 system, as shown in Fig. 4, and the signal recording process was initiated when all electrodes showed good quality (green light) in the software. In the software, electrodes with inferior quality (red light) or poor quality (orange light) were improved. This was done by applying conductive gel to the electrodes to increase conductivity or remove interfering hair strands. Once all electrodes reached 100% quality, the next step was taken. After the participants achieved concentration, the video clip was started, and EEG recording began. Simultaneously, the participants' button clicks indicating "yes" and "no" were also recorded. This allowed for repeating the experiment in case of missing or incorrect clicks. A 5% margin of error was disregarded for the clicks.

During the data collection phase, the stimuli used and the participants included in the study were described in detail in previous sections. In both experiments conducted for each participant, EEG data were collected during the presentation of video clips of the same length. Each video clip lasted 75 s, consisting of 1 s of the black screen and 2 s of stimulus image. The stimulus sequence diagram is shown in Fig. 5.

EEG signals obtained from the EEG device are converted to a floating-point value stored by the EmotivPro software. To measure negative values, the DC value of the signal is set at an average level of 4200 uV. Positive voltages occur above the average, while negative voltages occur below the

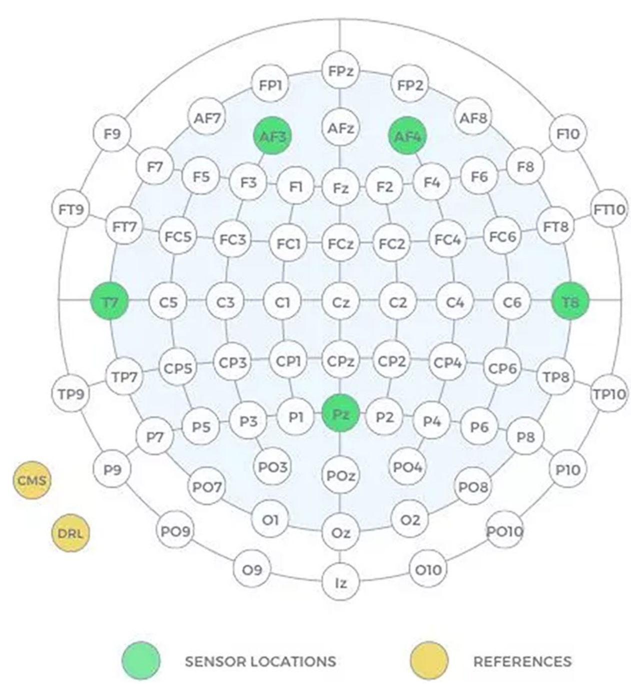


Fig. 4 Sensor and reference electrode locations [33]

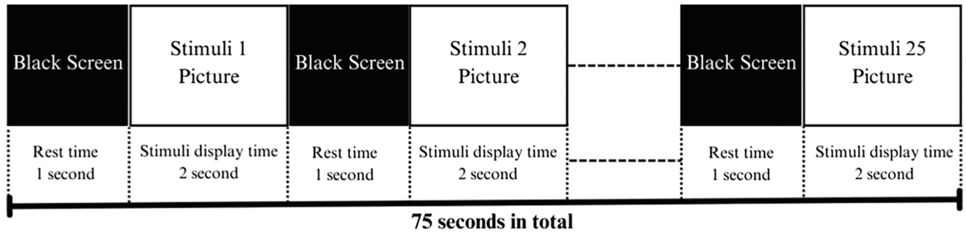


Fig. 5 Stimulus sequence diagram

average. The DC offset needs to be removed before any processing is performed on the signal. This is achieved by subtracting the average value from each data point [34]. Thus, the DC offset is eliminated.

Due to the sampling rate of the EEG device being 128 Hz, a total of 9600 data points were obtained during each experiment. After all data collection steps, the resulting LieWaves dataset has $27 \times 2 \times 5 \times 9600$ dimensions. Here, 27 represents the number of participants, 2 represents the number of experiments conducted, 5 represents the number of EEG channels used, and 9600 represents the number of samples. The dataset was collected from 27 participants using a 5-channel EEG device with fewer channels than those used in existing datasets.

3.2 Preprocessing

EEG data needs to undergo several preprocessing steps for better analysis. Preprocessing methods are applied to address artifacts or disturbances that may occur during signal acquisition, such as noise, muscle movements, electrode movements, eye movements, and heartbeats. Additionally, signals can be filtered within specific frequency ranges during preprocessing. This helps improve the results obtained from analyzing EEG data. The LieWaves dataset underwent preprocessing steps in this study, including applying a bandpass filter with a 0.5–45 Hz range. Subsequently, an artifact removal process was performed using the ATAR algorithm to eliminate artifacts.

3.2.1 Automatic and tunable artifact removal (ATAR)

The automatic and tunable artifact removal (ATAR) algorithm is based on wavelet packet decomposition (WPD). The ATAR method sets a threshold value and decomposes the EEG signal into wavelet packets. The threshold value is used to identify the components in the signal. Subsequently, all components below the threshold value are considered artifacts and removed. This process enables the fast and automatic detection of artifacts in the signal. The ATAR method is particularly effective for short-duration signals, making it a valuable technique for detecting and removing certain artifacts in EEG signals. Additionally, the ATAR method can be used with an adjustable wavelet filter to remove specific frequency components in the signal according to user preference. This feature allows the user to remove unwanted components and access desired components in the signal. The wavelet filter employs elimination, linear attenuation, and soft thresholding. The elimination filter eliminates wavelet components exceeding a specified θα threshold. In the linear attenuation filter, if the amplitude of a wavelet component is low, it is preserved but linearly attenuated beyond a specified θα threshold and eliminated beyond another θb threshold. In the soft thresholding filter, wavelet components exceeding the specified θa threshold are reduced to this threshold value instead of wholly eliminated [35].

In this study, the ATAR algorithm was used with default settings. A threshold value of 0.1 and soft thresholding as the wavelet filter were employed. Figure 6 depicts the signal status before and after applying ATAR on the 5 channels of the EEG dataset.

3.3 Overlapping sliding window (OSW)

It is ideal to have an adequate data size to process data using deep learning methods. Therefore, the OSW (over-sampling wavelets) method is used to augment the data on several samples [36–38]. After preprocessing, the OSW method was applied to increase the number of data samples. In this method, the time series data is divided into 384 segments. Then, a step process consisting of 32 filters is applied to obtain each segment. As a result, the available data size is increased. These values are selected based on experimental results. Figure 7 illustrates the schematic representation of the OSW method.

3.4 Feature extraction

A set of features is extracted from raw data to capture as much relevant information as possible [39]. Feature extraction is a crucial stage in the classification process, where non-artificial data is obtained by extracting and selecting only the relevant features based on the application. The main objective of feature extraction is to identify common patterns from EEG signals. The EEG signal needs to be analyzed in both the time and frequency domains to extract features. In this study, the methods of discrete wavelet transform (DWT) and fast Fourier transform (FFT) were applied, followed by

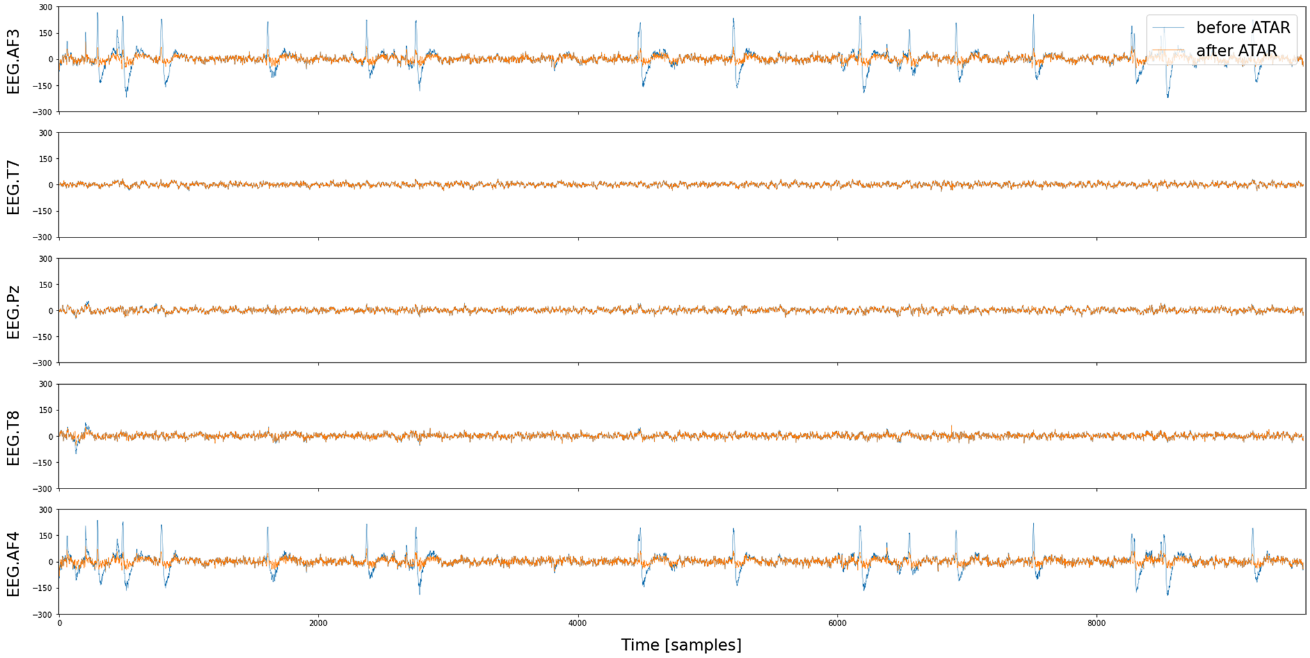


Fig. 6 A Signal display of 5 channels before and after ATAR is applied

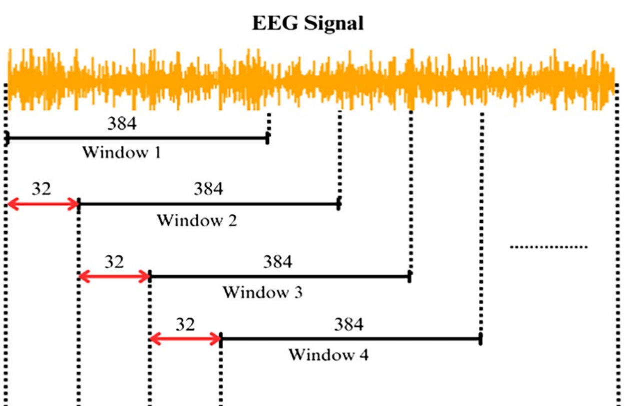


Fig. 7 Schematic representation of the OSW method

the application of statistical methods (SM) for each of them, resulting in feature vectors.

3.4.1 Statistical methods (İY)

After applying the DWT (discrete wavelet transform) and FFT (fast Fourier transform) methods during the feature extraction stage, various statistical methods can be applied to the obtained data. The statistical methods used in this study are as follows:

Minimum: The smallest value in the dataset.

Maximum: The largest value in the dataset.

Median: The median value in the dataset, calculated as shown in Eq. 1. Here, n represents the number of elements.

$$\text{Median} = \frac{(n+1)}{2} \tag{1}$$

Mean: The dataset's average is obtained by summing all values and dividing by the number of values, as shown in Eq. 2. Here, μ represents the mean of the signal data, x represents each signal data point, and n represents the number of elements.

$$\mu = \frac{\sum x}{n} \tag{2}$$

Energy: A measure of the magnitude of the EEG signal. It is obtained by summing the squares of the signals for each frequency band. Equation 3 represents the energy formula. X_i represents each signal data point, and n represents the number of elements.

$$\text{Energy}(E_i) = \sum_{i=1}^{n} |X_i|^2 \tag{3}$$

Relative energy: It is obtained by dividing the energy obtained for each band by the total energy. Equation 4 represents the total energy formula, and Eq. 5 represents the relative energy formula. Here, E_i represents the energy of each signal, and E_{total} represents the total energy.

$$\text{Energy}(E_{total}) = \sum_{i=1}^{n} E_i \tag{4}$$

$$\text{Energy}(E_{rel}) = \frac{E_i}{E_{total}} \tag{5}$$

Standard deviation: It indicates the amount of variation or dispersion from the mean in the dataset. A low standard deviation indicates that the data points are close to the mean, while a high standard deviation indicates that the data points are spread out over a wide range. It is calculated using the formula in Eq. 6. Here, X_{i} represents each signal data point, μ represents the mean of the signal data, and n represents the number of elements.

$$\sigma = \sqrt{\frac{1}{n-1}\sum\nolimits_{i=1}^{n}\left(X_{\mathrm{i}}-\mu\right)^{2}} \tag{6}$$

Variance: The variance of the dataset is equal to the square of the standard deviation. It is calculated using the formula in Eq. 7. Here, σ represents the standard deviation.

$$V = \sigma^2 \tag{7}$$

Skewness: It measures the similarity or asymmetry of the distribution. It can be positive or negative. It is calculated as shown in Eq. 8. In the equation, X_{i} represents the signal data point, μ represents the mean of the signal data, n represents the number of elements, and σ represents the standard deviation.

$$S = \frac{1}{(n-1)\sigma^3}\sum\nolimits_{i=1}^{n}\left(X_{\mathrm{i}}-\mu\right)^{3} \tag{8}$$

Entropy: Entropy is a measure used to quantify the uncertainty of a probability distribution. It is calculated using the formula shown in Eq. 9. Here, X_{i} represents a finite sequence of values, and pi represents the i element of the probability distribution.

$$H(x) = \sum\nolimits_{i=1}^{n}(p_i(\mathrm{x})log_2(p_i(x))) \tag{9}$$

Kurtosis: It measures the symmetry and distribution of EEG signals. The formula for kurtosis is given in Eq. 10. Here, n is the number of samples in the dataset, X_{i} represents the value of each sample, μ represents the mean of the dataset, and σ represents the standard deviation of the dataset.

$$K = \frac{1}{n}\frac{\sum_{i=1}^{n}(X_{\mathrm{i}}-\mu)}{\sigma^4} \tag{10}$$

Delta power ratio: It represents the ratio between the power in the delta frequency band (0.5–4 Hz) and the total power.

Theta power ratio: It represents the ratio between the power in the theta frequency band (4–7 Hz) and the total power.

Alpha power ratio: It represents the ratio between the power in the alpha frequency band (7–12 Hz) and the total power.

Beta power ratio: It represents the ratio between the power in the beta frequency band (12–30 Hz) and the total power.

Gamma power ratio: It represents the ratio between the power in the gamma frequency band (30–100 Hz) and the total power.

3.4.2 Discrete wavelet transform (DWT)

DWT (discrete wavelet transform) is the most suitable time–frequency analysis technique for non-stationary signals. It is chosen as a feature extraction method for these signals because essential features can be present in the time domain, frequency domain, or both. DWT also has other features, such as significant data reduction. It analyzes the signal in different frequency bands with different resolutions by decomposing it into approximation and detailed information [40].

To apply DWT, an efficient algorithm is designed using a series of quadrature mirror filters performed by low-pass (l) and high-pass (h) filters [39]. The cut-off frequency of the low-pass filter l(n) and high-pass filter h(n) equal half of the sampling frequency. In this study, with a sampling frequency of 128 Hz, when the low-pass filter is applied at the first level, signals in the frequency range of 0–64 Hz are obtained, and when the high-pass filter is applied, signals in the frequency range of 64–128 Hz are obtained. In the first step of DWT, the input signal is simultaneously passed through the low-pass and high-pass filters, and the outputs are called the approximation (A1) and detail (D1) coefficients at the first level. The same procedure is repeated to obtain the approximation and detail coefficients of the next level. At each decomposition step, the frequency resolution is doubled by filtering, and the time resolution is halved by downsampling [5, 41, 42]. The approximation and detail coefficients are obtained using low-pass and high-pass filters according to the formulas shown in Eqs. 11 and 12.

$$A[n] = (y \times X)[n] = \sum y[k] \times X[2n-k] \tag{11}$$

$$D[n] = (a \times X)[n] = \sum a[k] \times X[2n-k] \tag{12}$$

Here, $x[n]$ is the original signal or dataset, $y[k]$ represents the coefficients of the high-pass filter, $a[k]$ represents the coefficients of the low-pass filter, $A[n]$ is the output of the approximation coefficients, $D[n]$ is the output of the detail coefficients, n is the index value, and k is the index value of the filter coefficients. Figure 8 shows the schematic view of DWT. In this case, a fourth-level decomposition using a fourth-order Daubechies wavelet filter is performed, and the obtained D1, D2, D3, D4, and A4 values represent the sub-bands.

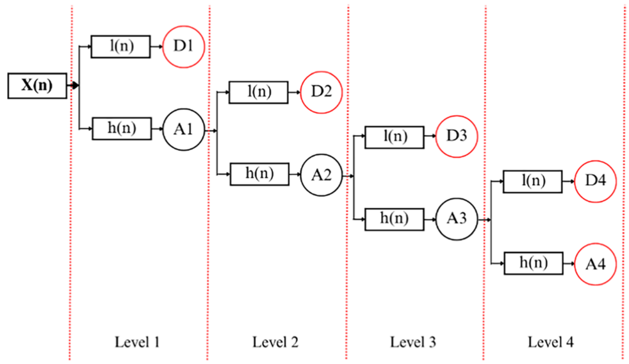


Fig. 8 DWT subcomponent decomposition diagram

After applying the DWT method, the resulting subcomponents were further extended by using statistical methods. Following the application of the DWT method, statistical methods such as minimum, maximum, median, energy, relative energy, mean, variance, standard deviation, and skewness were individually applied to each subcomponent. These statistical methods are described in Section 3.4.1. An overall feature matrix of size 250 ($5 \times 5 \times 10$) was obtained for analysis across all channels. For channel-based analysis, feature matrices of size 50 ($1 \times 5 \times 10$) were created.

3.4.3 Fast Fourier transform (FFT)

FFT (fast Fourier transform) is a fast method for discrete Fourier transformation. In feature extraction of EEG signals, it transforms the EEG signal from the time domain to the frequency domain and performs spectrum analysis or calculates power spectral density [43]. This method allows for analyzing EEG signals in the frequency and time domains. It has a good performance on stationary data and linear systems. However, it does not work well with non-stationary data and cannot perform calculations in both time and frequency domains. FFT is a mathematical transformation method for transforming a signal into the frequency domain. The basic formula of FFT is used to represent a given signal in the time domain as a representation in the frequency domain. Equation 13 represents the mathematical representation of FFT.

$$X[k] = \sum x[n] \times e^{\left(\frac{-i \times 2\pi \times k \times n}{N}\right)} \tag{13}$$

Here, $X[k]$ represents the FFT output for the kth frequency component, $x[n]$ represents the nth time sample, e represents Euler's number ($e \approx 2.71828$), i represents the imaginary unit ($\sqrt{(-1)}$), π represents the pi number ($\pi \approx 3.14159$), k represents the index of the frequency component (from 0 to $N-1$), n represents the index of the time sample (from 0 to $N-1$), and N represents the total number of samples. The FFT formula uses the given samples in the time domain to calculate the signal's frequency components. Each frequency component is multiplied by the time sample and includes a complex term expressed as the power of Euler. As a result, the frequency components $X[k]$ are obtained.

In the FFT process, the power spectral density is calculated. Different features can be extracted from the power spectral density. In this study, after applying the FFT method, features such as mean, standard deviation, skewness, kurtosis, entropy, delta band power ratio, alpha band power ratio, theta band power ratio, beta band power ratio, and gamma band power ratio were extracted from the power spectral density to form a feature vector. The statistical methods described in Section 3.4.1 were applied in this stage. An overall feature matrix of size 50 (5×10) was obtained for the analysis across all channels. For channel-based analysis, feature matrices of size 10 (1×10) were created.

3.5 Classification

The feature vectors obtained after the feature extraction stage were used in the classification process. In the classification stage, CNN, LSTM, and CNNLSTM models, which combine both methods, were employed.

3.5.1 Convolutional Neural Networks (CNN)

Convolutional Neural Network (CNN) is a deep learning architecture commonly used for image classification. In classification tasks involving images and videos, 2D-CNN is typically employed. However, in classifying time series data such as EEG signals, 1D-CNN is more frequently used. Using CNN architecture alongside traditional methods has become widespread in EEG signal classification and has shown promising results. The CNN architecture automatically performs feature extraction in the classification process of EEG signals. It consists of multiple hidden layers, including convolutional, ReLU, pooling, and fully connected layers. The convolutional and pooling layers are employed for feature extraction, while the fully connected layer serves as a classifier. Since real-world data is often non-linear, CNN can learn non-linear patterns. The ReLU layer is used to transform the linear operation of the convolutional layer into a non-linear one [44–46]. In this study, the architecture of the implemented CNN model is illustrated in Fig. 9. The parameter values used were determined through trial and error. The best results were obtained with the architecture shown in the figure and the corresponding parameter values.

3.5.2 Long-Short Term Memory (LSTM)

The Long-Short Term Memory (LSTM) algorithm is a recurrent neural networks (RNN) subcategory. RNN has achieved significant success in various fields, including

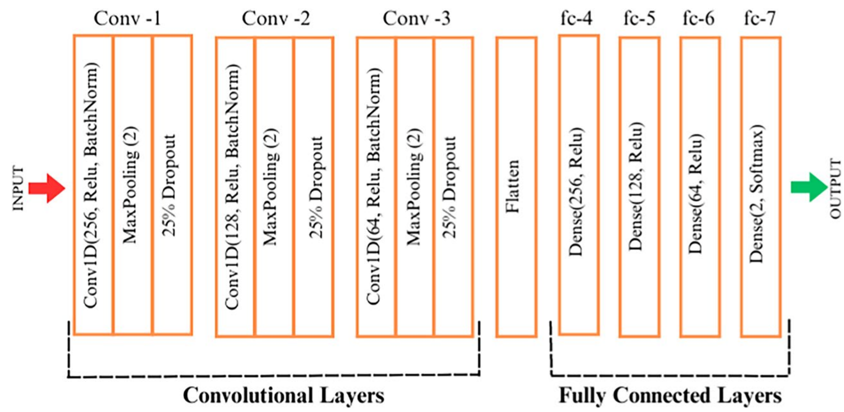


Fig. 9 CNN architecture was applied in the study

time series analysis. In RNN structures, there is a connection between the input and output. This means that the output at a given step influences the input of the next step. Unlike classical deep learning algorithms, RNN does not discard preprocessed data but retains it for subsequent steps. RNN attempts to store the data in its memory. However, in the case of large-scale data, memory limitations may prevent obtaining reliable results, which poses a significant challenge for RNN [45]. To overcome this problem, the LSTM algorithm was developed in subsequent studies [47]. This allows for more extended memory and the retention of more historical data. RNN usually has a single layer, whereas LSTM has a four-layered structure. The LSTM structure consists of a cell state, input gate, forget gate, and output gate. The cell state can be described as a pathway carrying information throughout the cells, serving as the memory of the network. The gates compress the incoming data between 0 and 1 using a sigmoid function. If the function's output is 0, the data is forgotten; if the output is 1, the data is propagated with the cell state [10]. Figure 10 illustrates the cell structure of the LSTM algorithm.

The architecture of the implemented LSTM model in this study is shown in Fig. 11. The parameter values used were determined through a trial-and-error approach. The best results were obtained with the architecture depicted in the figure, using the applied parameters and values.

3.5.3 CNNLSTM model

CNNLSTM (Convolutional Neural Network-Long-Short Term Memory) is an artificial neural network model. This model is mainly used to classify and predict time series data. CNNLSTM combines the features of both CNN and LSTM models. CNN is a Convolutional Neural Network used for extracting features from data. LSTM, on the other hand, is used for analyzing long-term dependencies in time series data. The CNNLSTM model first sends the data to the Convolutional Neural Network for feature extraction. Then, it

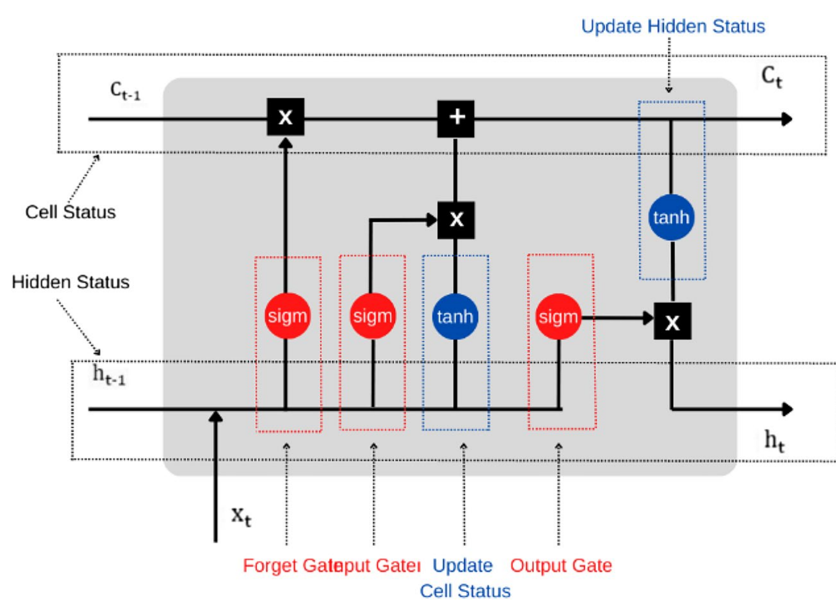


Fig. 10 The Long-Short Term Memory (LSTM) cell structure [10]

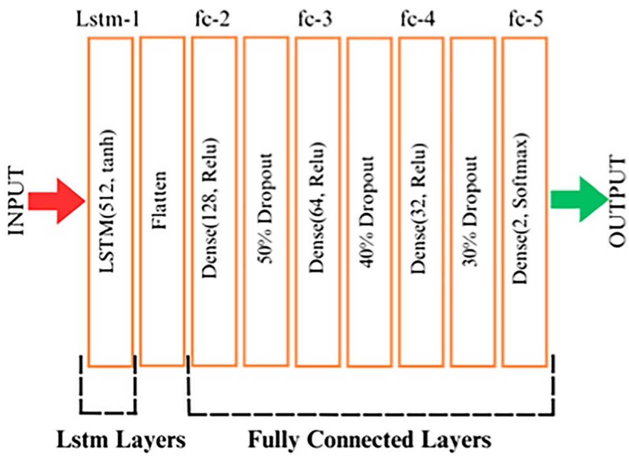


Fig. 11 LSTM architecture was applied in the study

passes the features to LSTM layers for further processing. This allows the model to be used for feature extraction and analysis of time series dependencies. The architecture of the implemented CNNLSTM model in this study is shown in Fig. 12. The parameter values used were determined through a trial-and-error approach. The best results were obtained with the architecture depicted in the figure, using the applied parameters and values.

4 Results and discussion

This study created a dataset named LieWaves for lie detection using EEG signals, which consisted of 5 EEG channels. Analyses were performed on this dataset for lie detection. The processing and analysis stages of EEG signals were applied step by step. The steps for creating the dataset were explained individually. After obtaining the dataset, a band-pass filter was applied to the 0.5–45 Hz EEG data. Then, the ATAR algorithm was used to remove artifacts in the EEG signals. After this stage, whole-channel and channel-based analyses were conducted on the dataset. The OSW method was used to augment the data in the dataset to increase the analysis results. This allowed for classification with a more significant number of data samples and resulted in better results. For the classification of the data, feature extraction was performed first. In this stage, the DWT and FFT methods were used, and the features obtained from each method were used separately in the classification step. The obtained feature vectors were subjected to classification using the CNN, LSTM, and CNNLSTM models separately. Before the classification stage, the data was divided into training, validation, and test sets. The total data was set to be 80% for training, 10% for validation, and 10% for testing. Subsequently, normalization was applied to all the data. Normalization is scaling the data to a standard scale from − 1 to 1, allowing for comparing different data types. The iteration count was 100, and the batch size was 100 for all three models. Other values were also tried, but the best results were obtained with these values. The results obtained after applying all the methods and models are shown in Table 1.

The values highlighted in bold in Table 1 represent the best results obtained in the study. For both all channels and channel-based evaluations, the best outcomes obtained for each feature extraction method and each deep learning model are indicated in bold. According to the findings shown in Table 1, the highest accuracy rate of 99.88% was achieved when the DWT + LSTM method was applied to the data that included all channels. When examining the data, it can be observed that the best performances were obtained from the data on which the DWT method was applied. The comparison between the DWT and FFT methods applied in the feature extraction stage is evident from these results. The results of the classification models, on the other hand, show very close values to each other. However, when looking at the average values, the highest performance values are observed to be obtained with the LSTM model.

When the feature vector obtained from applying the DWT method to the data that included all channels was classified using the CNN, LSTM, and CNNLSTM models, accuracy rates of 99.81%, 99.88%, and 98.76% were achieved,

Fig. 12 CNNLSTM architecture was implemented in the study

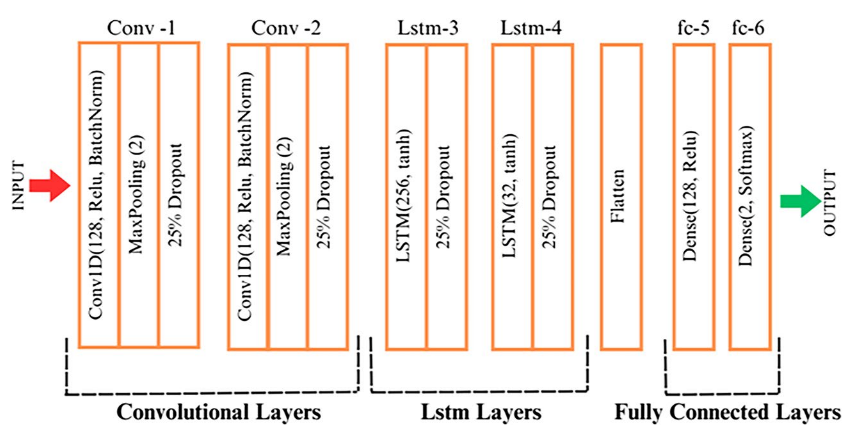


Table 1 Results obtained as a result of classification

	DWT			FFT		
	CNN	LSTM	CNNLSTM	CNN	LSTM	CNNLSTM
All channels	99.81	**99.88**	98.76	98.52	98.95	97.65
AF3	95.61	95.49	95.30	**66.07**	61.00	62.55
T7	95.61	95.49	94.56	69.72	66.93	**71.51**
Pz	96.54	96.66	96.29	**67.92**	64.15	67.68
T8	**97.03**	96.91	**96.91**	**73.79**	66.32	66.87
AF4	95.98	**97.16**	95.36	67.24	61.87	**68.11**
Average	96.77	**96.93**	96.20	**73.88**	69.87	72.39

respectively. When the feature vectors obtained from channel-based data were classified with these models, the highest accuracy rate of 97.16% was obtained when the LSTM model was applied to the AF4 channel data. When looking at the other channels, the CNN applied to the AF3, T7, and T8 channels achieved the highest performances with accuracy rates of 95.61%, 95.61%, and 97.03%, respectively. The data from the Pz channel achieved the highest performance with an accuracy rate of 96.66% when the LSTM model was applied. According to the averages of the classification results obtained on the data using the DWT method, the best classification performance of 96.93% was achieved with the LSTM model. Figure 13 provides the training-validation accuracy graph and confusion matrix obtained when the DWT + LSTM methods were applied to all channels.

When the FFT method was applied to the data that included all channels, the feature vector obtained was classified using the CNN, LSTM, and CNNLSTM models, resulting in accuracy rates of 98.52%, 98.95%, and 97.65%, respectively. When the feature vectors obtained from channel-based data using the FFT method were classified with these models, the highest accuracy rate of 73.79% was achieved when the CNN model was applied to the T8 channel data. When looking at the other channels, the CNN applied to the AF3 and Pz channels yielded accuracy rates of 66.07% and 67.92%, respectively. The CNNLSTM applied to the T7 and AF4 channels resulted in accuracy rates of 71.51% and 68.11%, respectively. According to the averages of the classification results obtained on the data using the FFT method, the best classification performance of 73.88% was achieved with the CNN model. Figure 14 provides the training-validation accuracy values and confusion matrix graphs obtained when the LSTM method, which gave the best results among the models using the FFT method, was applied to all channels.

Figure 15 presents the overall results graph, showing all the results obtained. All the results indicate the success of the study. When comparing the DWT and FFT methods used in the feature extraction stage, the DWT method yields better results in all models. Although the FFT method yielded high results in the analysis performed on all channels, it produced lower results in channel-based analyses. The higher number of channels increases the dimension of the feature vector. In the case of single-channel analysis, the feature vector size is expected to remain low, leading to lower results. Despite using different features and increasing the number of features, the best result was still obtained from these employed features.

When comparing the CNN, LSTM, and CNNLSTM models used in the classification stage, it can be observed that all the results are very close to each other. However, generally, the LSTM model performed better when used alone. In some channel-based analyses, the CNN model showed slight superiority, while the CNNLSTM model exhibited a slight

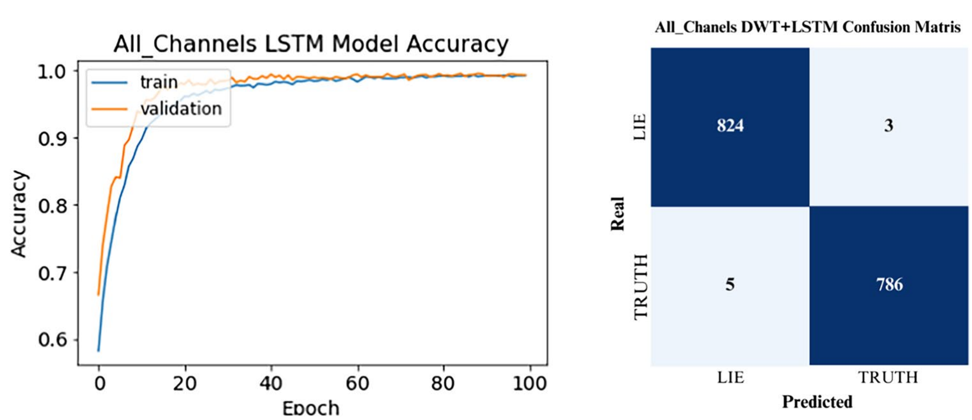


Fig. 13 Accuracy graph and confusion matrix of DWT + LSTM methods applied on all channels

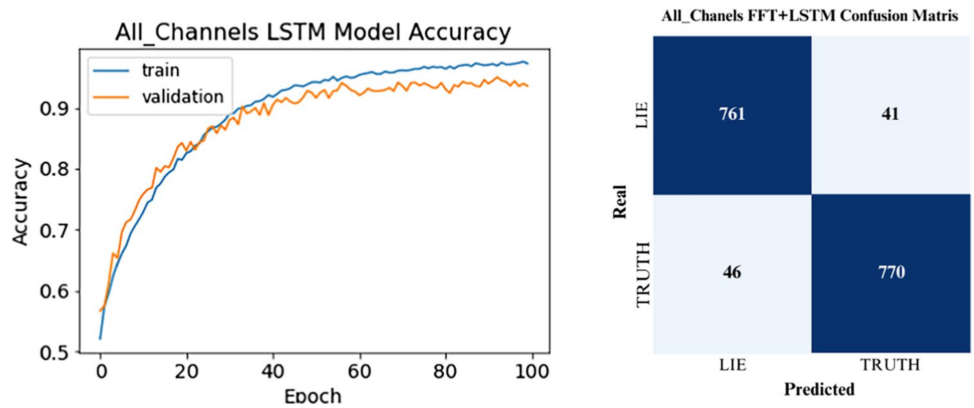
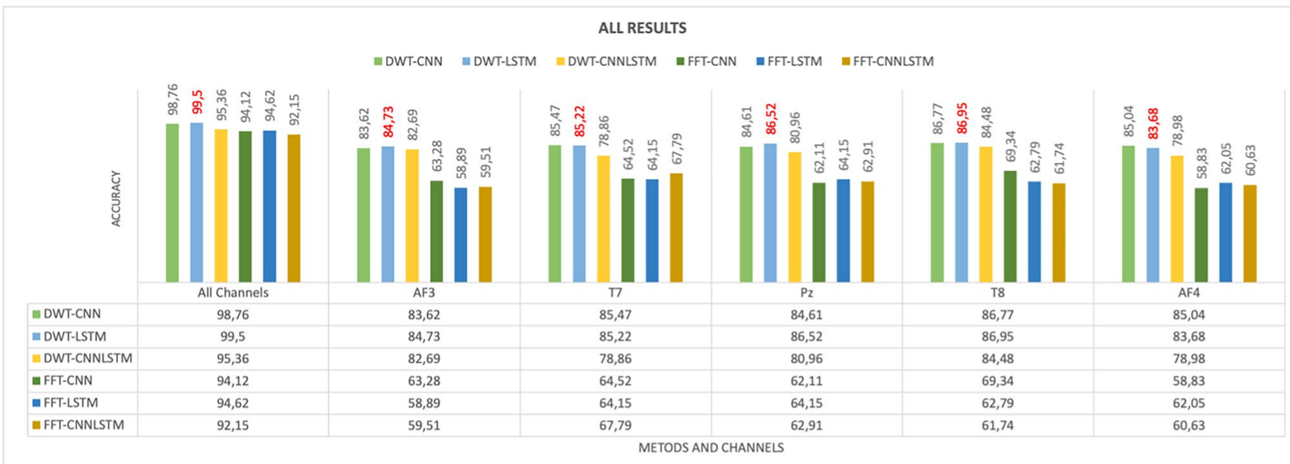


Fig. 14 Accuracy graph and confusion matrix of FFT + LSTM methods applied on all channels



	All Channels	AF3	T7	Pz	T8	AF4
DWT-CNN	98,76	83,62	85,47	84,61	86,77	85,04
DWT-LSTM	99,5	84,73	85,22	86,52	86,95	83,68
DWT-CNNLSTM	95,36	82,69	78,86	80,96	84,48	78,98
FFT-CNN	94,12	63,28	64,52	62,11	69,34	58,83
FFT-LSTM	94,62	58,89	64,15	64,15	62,79	62,05
FFT-CNNLSTM	92,15	59,51	67,79	62,91	61,74	60,63

Fig. 15 Graph showing all results

advantage in others. In conclusion, all three models yielded successful accuracy values.

Looking at the channel-based data analysis, the results are close to each other. Since each channel represents specific brain regions, the study aimed to determine which brain region is more involved in deception. The fact that very close values are obtained indicates that all points of the brain are interconnected. Although it is mentioned in the literature that each brain region has its own operations, it is known that each region works by influencing each other [48]. In this study, when examining the channel-based results, the data from the T7 and T8 channels, which are electrodes used in the temporal region, produce higher accuracy rates in almost all methods. This is because the temporal region is responsible for processes such as memory and storing visual and auditory data in memory [48, 49]. Since visual stimuli were used during the dataset acquisition, the participants remembered these stimuli using memory techniques. Thus, more intense signals were observed in the temporal region, and the data obtained from there yielded more accurate values.

Table 2 presents studies conducted in the literature on deception detection using EEG signals, along with the datasets utilized in this study, applied methods, and obtained results.

Table 2 provides information on the datasets used in this study and several studies in the literature on deception detection using EEG signals. The table includes details such as the dataset names, the number of participants in each dataset, the number of channels in the EEG device used, the applied feature extraction methods, the classification methods employed, and the results obtained. The purpose of preparing this table is not only to compare the studies but also to highlight the unique aspects of this study compared to the methods applied in the literature. Upon reviewing the information, it is evident that the results obtained in this study are higher than those of other EEG deception detection studies in the literature. This outcome demonstrates the success of the study. Several factors contribute to the effectiveness of the proposed study, including the following:

Table 2 Information regarding this study and literature studies

References	Datasets	Number of participants	Number of channels	Feature extraction	Classification	Accuracy result (%)
[23]	Dryad.2qc64	30	12	DWT	SVM	96.05
[27]	Private	12	5	DWT	ANFIS	64.27
[28]	Private	10	16	EMD	SCM, KNN, QDA, DT	SVM: 81.71
[29]	Private	33	9	EMD	SVM	99.44
[30]	Dryad.2qc64	30	12	ICA	SVM, LDA, KNN, BPNN	KNN: 97.9
[31]	Private	80	1	-	SVM, DF, NN	DF: 99.2
[24]	Dryad.2qc64	30	12	-	CNN	84.44
[19]	Bag-Of-Lies	35	13	-	RF, EEGNet, MLP	RF: 58.71
[22]	Bag-Of-Lies	35	13	Time Series	LSTM	82.4
[20]	Bag-Of-Lies	35	13	WPT	DNN	95
This study	**LieWaves**	**27**	**5**	**DWT, FFT**	**CNN, LSTM, CNNLSTM**	**DWT + LSTM, 99.88; DWT + CNN, 99.81**

- The DWT method used in the feature extraction stage is a commonly employed technique in EEG signal processing due to its effectiveness in extracting valuable features from signals. Classifying the feature vector obtained from this method with a deep learning algorithm contributes to the improved performance rate.
- The ATAR algorithm, employed in the preprocessing stage of the study, is not widely used, yet it has demonstrated higher results compared to other methods. This superiority is attributed to the algorithm's better removal of unnecessary artifacts from the signals.
- Data augmentation was performed using the OSW method in the proposed study, contributing to the high performance achieved during the training stage.
- Different methods and models were employed to enhance efficiency, and parameters were fine-tuned through numerous trial-and-error iterations to ensure that the results obtained after each method were close to each other. This approach led to the selection of parameters yielding the best results.
- The usage of the CNNLSTM model in deception detection with EEG signals has yet to be encountered in the literature. Although introduced for the first time in this study, the results were similar to those of the LSTM model. Nevertheless, the results indicated that this model is still viable and produces excellent results.

5 Conclusion

The proposed study provides a new dataset named LieWaves for deception detection using EEG signals. This dataset was created through experiments with 27 subjects using prayer beads as visual stimuli. Each subject participated in two experiments, assuming the roles of a liar and a truth-teller. In each experiment, after taking two beads from a box containing five beads, the subject, as a liar, pressed the "No" button for the image of the bead seen on the screen and the "Yes" button for the bead not taken. In the truth-teller role, the subject was instructed to respond correctly to all images. EEG data were collected from each subject during truthful and deceptive statements, creating a dataset. The Emotiv Insight device with 5 channels, not commonly used in EEG data collection, was chosen. After obtaining the dataset, DC offset removal was applied, followed by a 0.5–45 Hz bandpass filter. Subsequently, the ATAR algorithm was employed to eliminate signal artifacts during data acquisition for further analysis. The OWS method was applied for data augmentation in the dataset, increasing its size for classification. After this process, analyses were conducted on all channels and channel-specific data. The study used DWT and FFT methods for feature extraction, and the obtained feature vectors were classified using CNN, LSTM, and CNNLSTM models. The highest accuracy, 99.88%, was achieved by applying DWT + LSTM methods on all channels. The results of the study are presented in Table 1. Considering all these results, the success of the study is evident. The values contributed to the literature and society by this study are as follows:

- Considering the results obtained in this study and those in similar studies [19, 20, 22–24, 27–31], it is evident that deception detection from EEG signals is possible, and high accuracy rates can be achieved.
- A new dataset named LieWaves was created for deception detection using EEG signals. Due to the limited availability of accessible datasets, studies in this field are constrained. This dataset will contribute to new research,

providing diversity with different procedures, subjects, and channel numbers.

- Results obtained for deception detection using the LieWaves dataset with fewer channels were higher than those obtained from other multi-channel datasets. This indicates the feasibility of deception detection from EEG signals with fewer channels, enabling cost-effective implementations.
- The methods and models used together in the study are less common in the examined deception detection studies, introducing innovation to the methods used for deception detection from EEG signals.
- The ATAR algorithm, utilized for noise removal in EEG signals, is not commonly encountered but demonstrated success in this study.
- Excellent results were achieved using DWT and LSTM models together.
- The CNNLSTM model, not found in the examined studies, was introduced in this study and showed high success rates. The study proved the potential applicability and effectiveness of the CNNLSTM model in future similar studies.
- The materials and methods used in the study are essential for providing insights into future studies.
- When evaluating the societal contribution of this study, it is foreseen that a deception detection system can be developed in the future using the developed model in this study. This system could benefit society by being used in areas where deception detection is crucial.

In addition to these contributions, there are some shortcomings or points to consider in the future:

- Relying solely on EEG signals for deception detection in such a critical subject as deception may raise questions in contemporary times. Including other data types in the dataset, such as EKG, pulse, body response, and facial analysis, can result in a more complex but reliable dataset. Training a model with this dataset could be practically used.
- One of the study's most critical aspects is whether it applies in practice. The process of collecting EEG signals is a laborious process, which may complicate practical applications. To address this issue, faster data acquisition steps can be applied using recently developed user-friendly EEG devices.
- In the future, during the collection of EEG signals for a new dataset, different types of stimuli (e.g., question–answer), different types of subjects, and different channel numbers (single channel) can be used. In particular, achieving high performance from a dataset created with user-friendly EEG devices will further emphasize the importance of EEG signals in deception detection.
- In addition to this study, conducting studies applying different methods to the LieWaves dataset to strengthen its performance would be beneficial.
- Visualization of non-image data and classification using a 2D-CNN model [20, 50] can be applied to visualize EEG signals, and these visual data can be classified using a 2D-CNN model. A study [20] has already conducted a similar approach, achieving high performance. Additionally, it is one of the studies that yielded the highest results among those using the Bag-of-Lies dataset. Applying this method to the LieWaves dataset could lead to even higher results.

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Data availability Researchers can access the dataset at https://data.mendeley.com/datasets/5gzxb2bzs2/1.

Declarations

Ethical approval The EEG recordings obtained for creating the LieWaves dataset and the methods applied to the participants during the signal acquisition stage were conducted in full compliance with the ethical approval granted by the ethics committee.

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